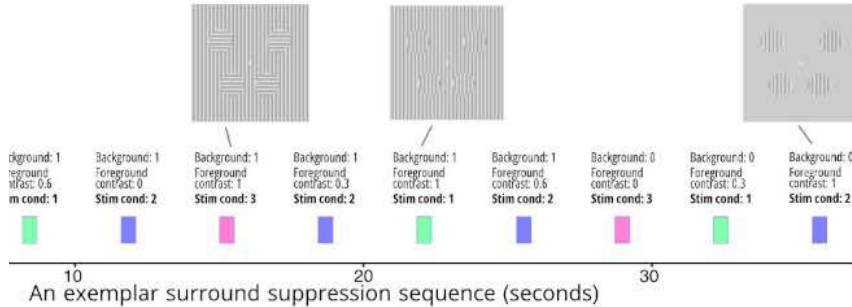
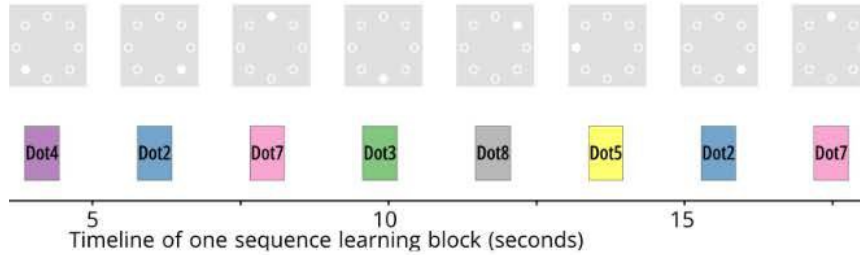


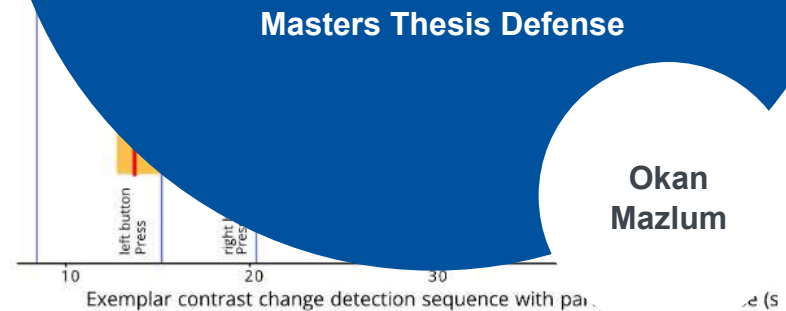
University of Stuttgart
Germany



Insights from a very large EEG+Eyetracking dataset

Masters Thesis Defense

Okan
Mazlum

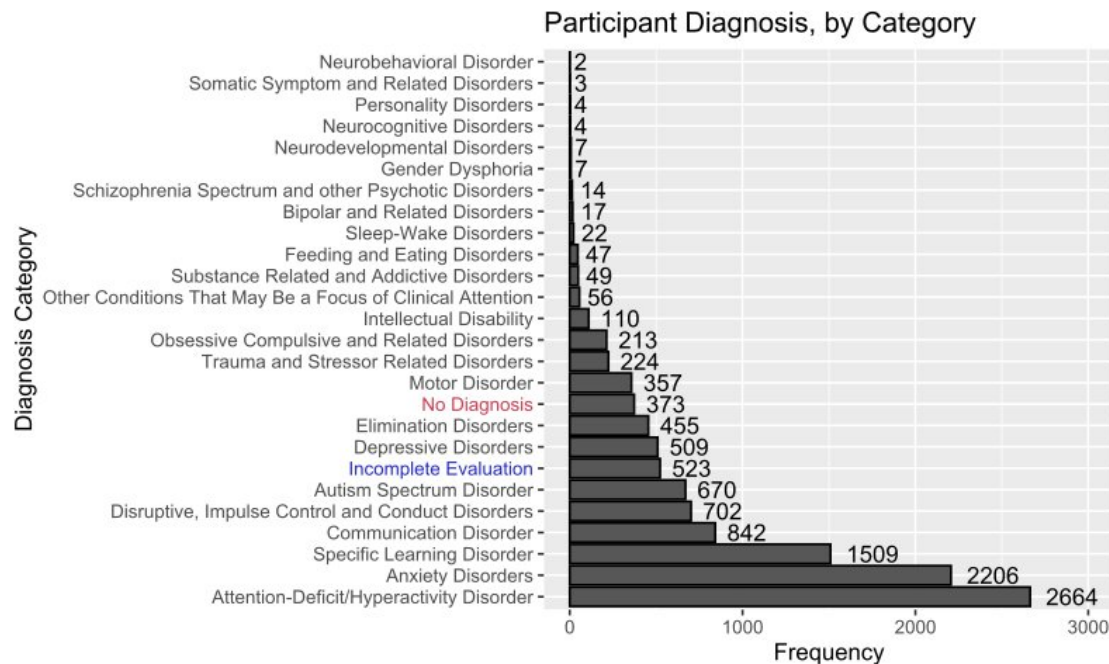


Structure

- **The Dataset**
- Intro to FRPs
- Research Questions
- Dataset Preparation
- Subject Curation
- Analysis Results
- Takeaways

The Healthy Brain Network (HBN) dataset

- Ongoing initiative to capture data from at least 10,000 adolescents
- Currently 11 Releases with 4000+ subjects aged 5-21

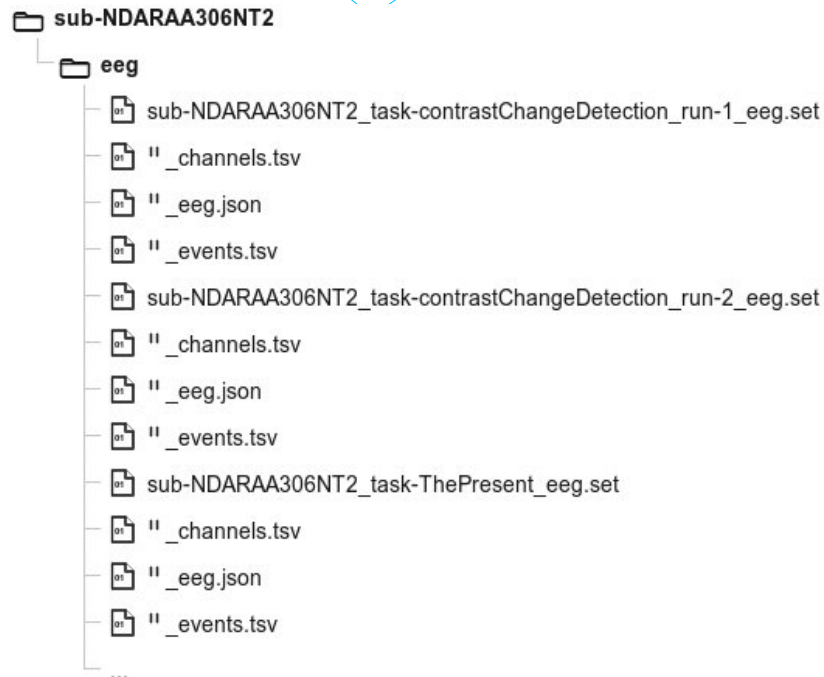


The Healthy Brain Network (HBN) dataset

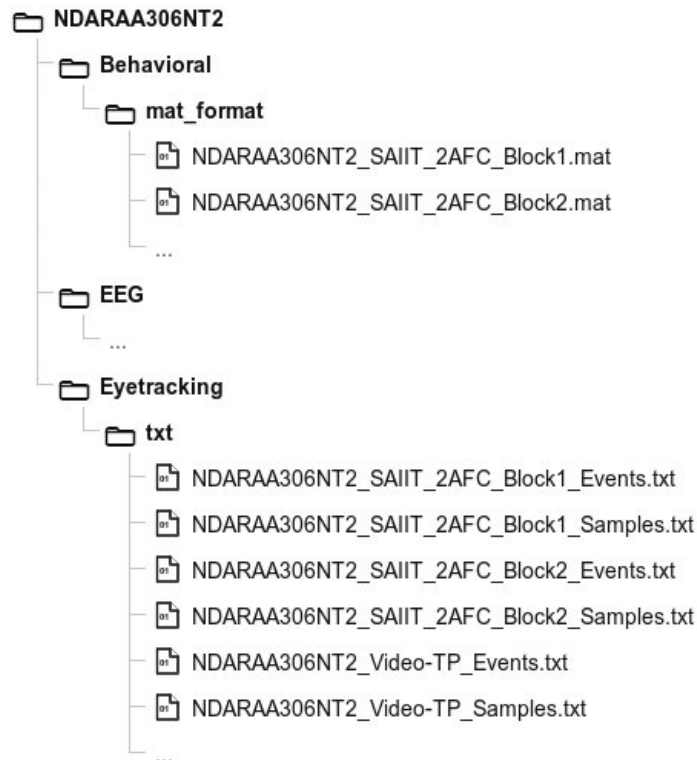
Paradigm name	Active?	Free viewing?	# of recordings	Mean duration
Resting State	<i>not used in this thesis</i>	<i>not used in this thesis</i>	<i>not used in this thesis</i>	<i>not used in this thesis</i>
Surround Suppression				
Contrast Change				
Sequence Learning				
Symbol Search	Yes	Yes	1282	03:42
Naturalistic Viewing	No	Yes	1841×4	03:03

The Healthy Brain Network (HBN) dataset

HBN-EEG



HBN



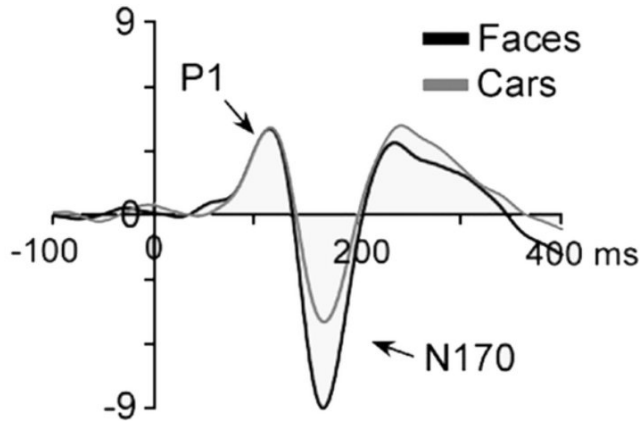
Structure

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FRPs

The relationship between EEG + ET

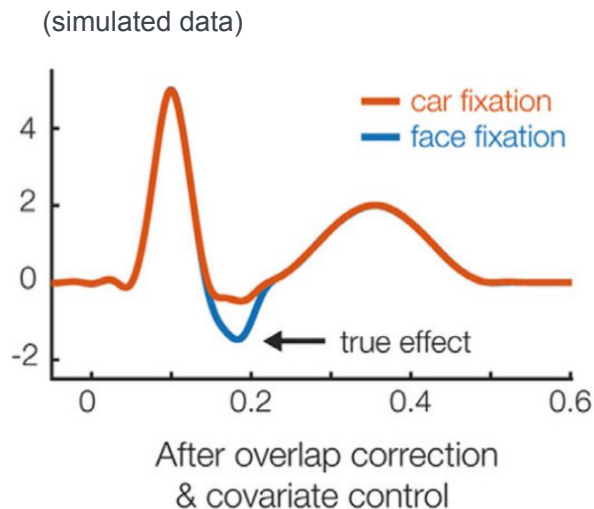
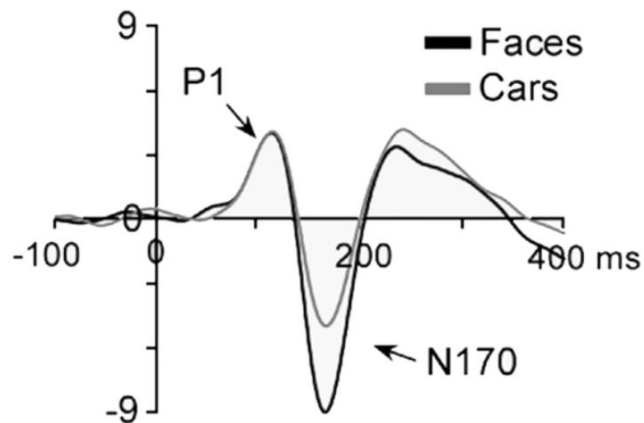
- Traditionally, EEG experiments minimize eye movements
 - Interest lies in event-related potentials (ERPs)



FRPs

The relationship between EEG + ET

- Traditionally, EEG experiments minimize eye movements
 - Interest lies in event-related potentials (ERPs)
- Eye movements do produce a meaningful neural response
 - Fixation-related potentials (FRPs)



Regression Model

Symbol Search formula

FRP $\sim$ 1

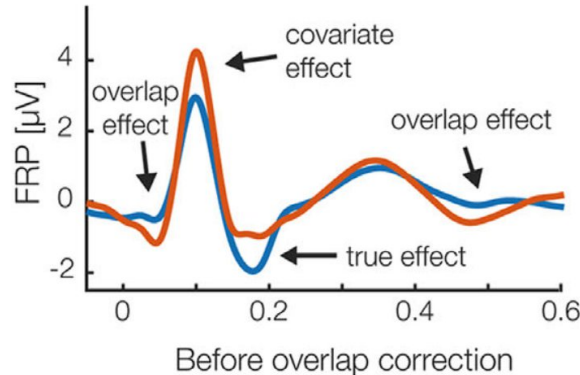
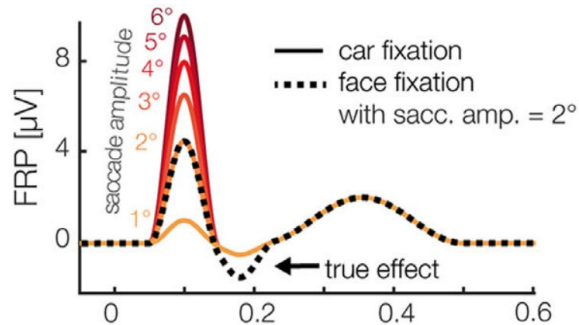
+spl(sacc_amplitude, 5)

+spl(fixation_position_x, 5)

+spl(fixation_position_y, 5)

⋮

- Intercept term
- Nonlinear effect of saccade amplitude
- Nonlinear effect of fixation position



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Research Questions

- Can HBN-EEG and eye-tracking data be transformed into a unified, usable resource?

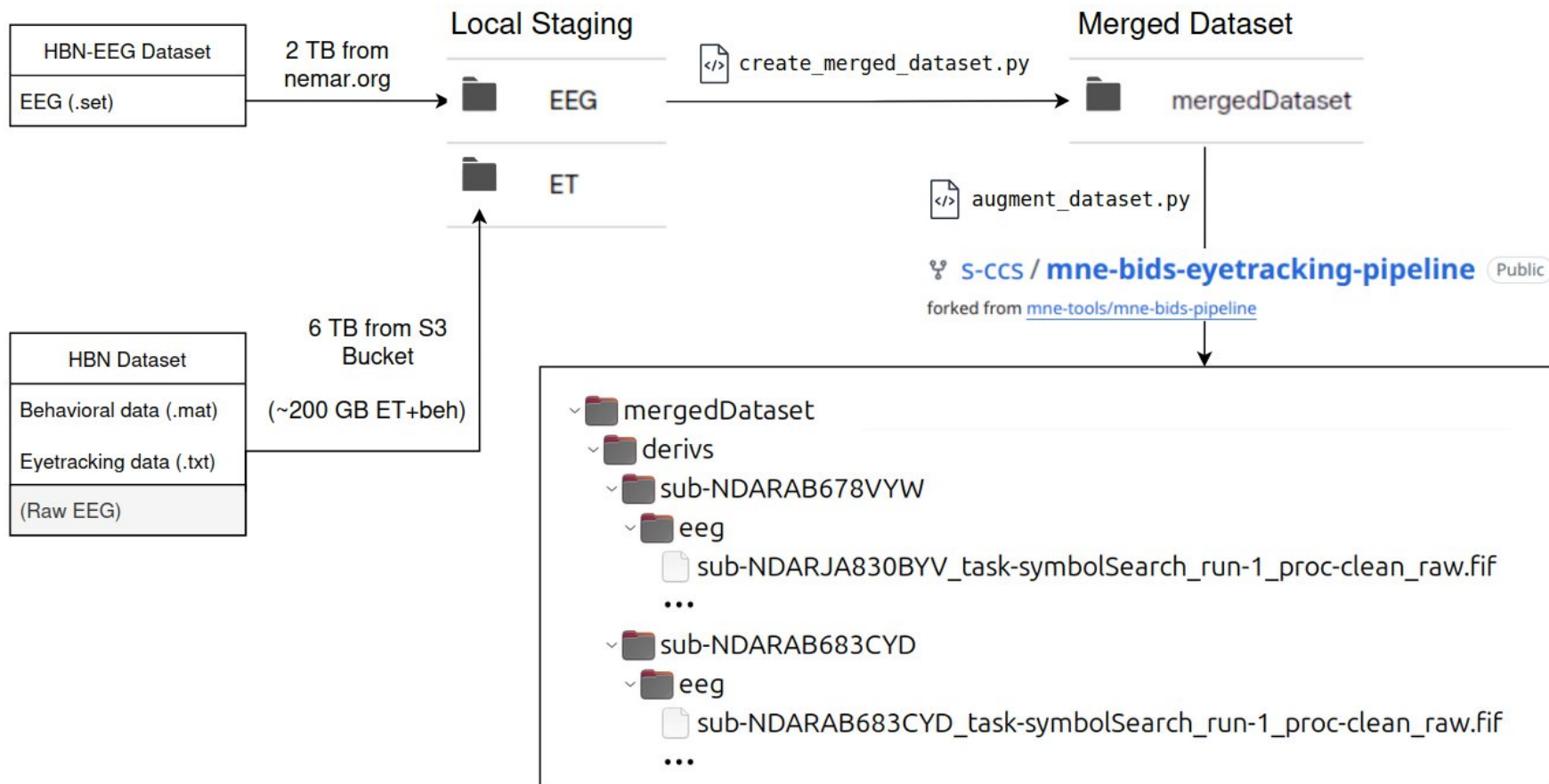
Research Questions

- Can HBN-EEG and eye-tracking data be transformed into a unified, usable resource?
- Can the scale and diversity of HBN provide new insight through FRP analysis?

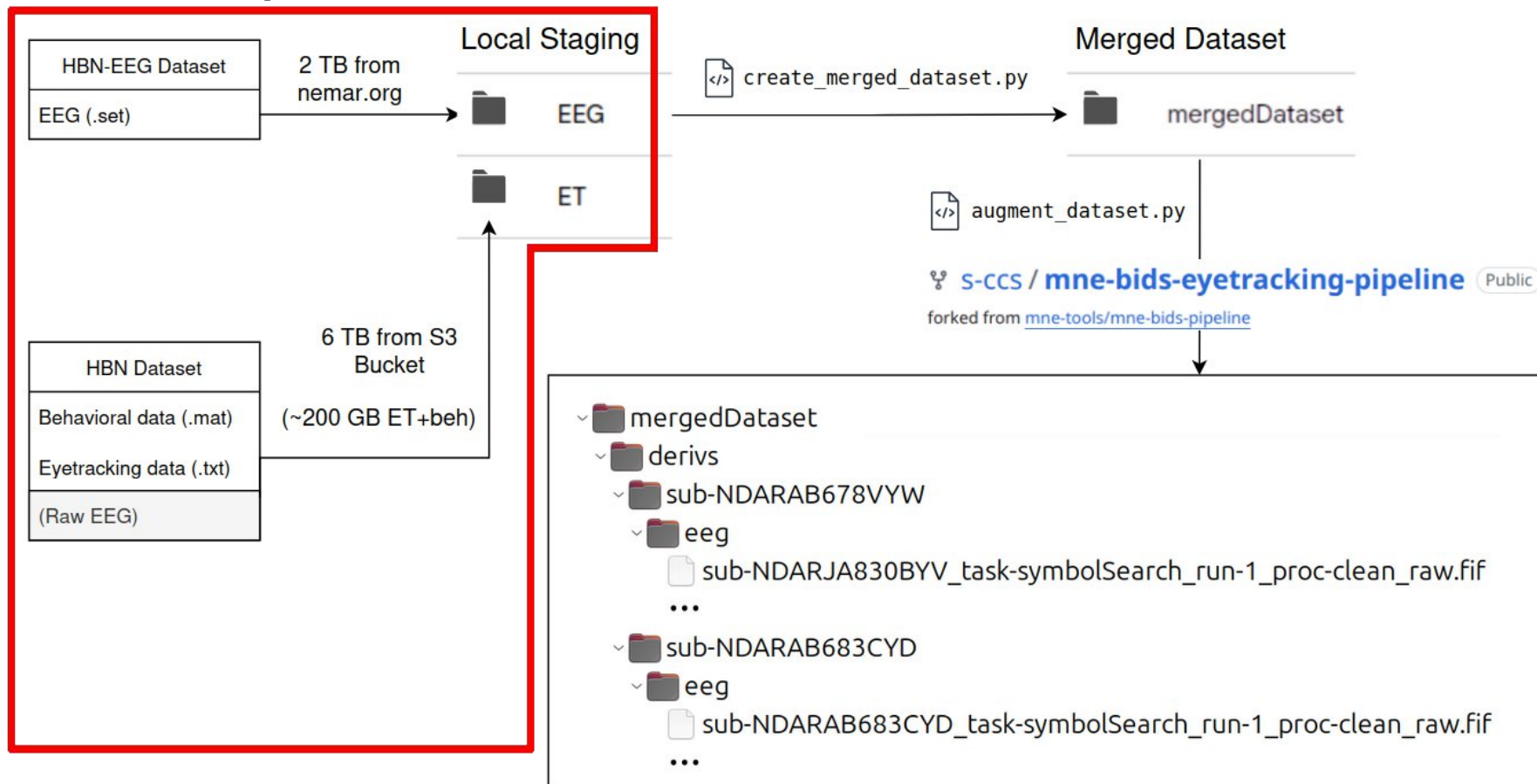
Structure

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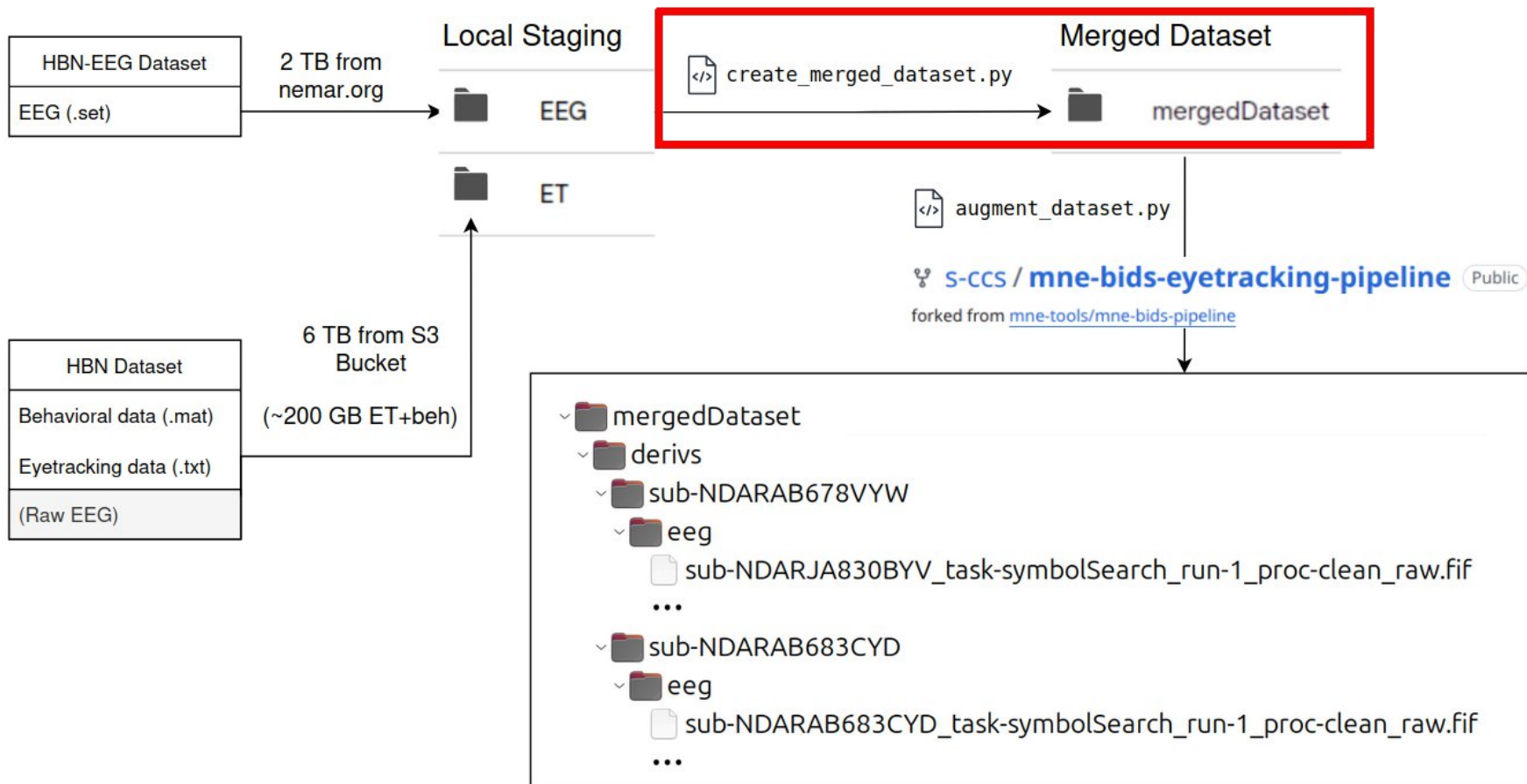
Dataset Preparation



Dataset Preparation



Dataset Preparation



Dataset Preparation



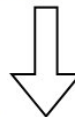
- Combine root-level metadata and quality tables

Dataset Preparation



- Combine root-level metadata and quality tables
- Map ET recordings to correct tasks

```
*__NODOOR_SAIIT_2AFC_Block1_Events.txt  
5008455_Video4_Events.txt  
*__Video-DM_Events.txt  
*_resting_Events.txt.save  
*__WISC_ProcSpeed_Events.txt
```



```
*_task-contrastChangeDetection_run-1_et_Events.txt  
*_task-ThePresent_et_Events.txt  
*_task-DespicableMe_et_Events.txt  
*_task-RestingState_et_Events.txt  
*_task-symbolSearch_et_Events.txt
```

Dataset Preparation



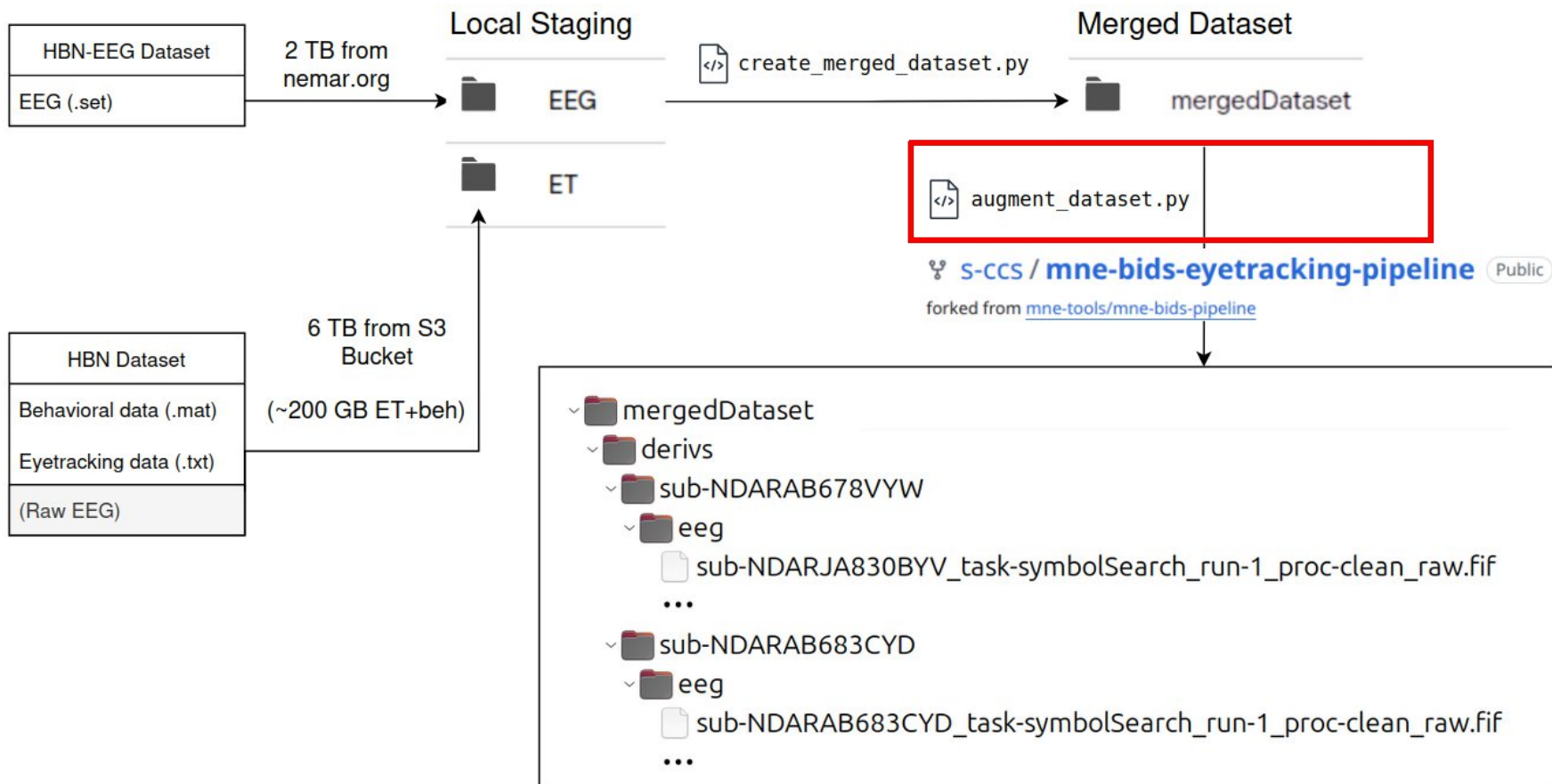
- Combine root-level metadata and quality tables
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- Fix `*_channels.tsv` (units → uV, type → EEG / MISC)

Dataset Preparation



- Combine root-level metadata and quality tables
- Map ET recordings to correct tasks
- Fix `*_channels.tsv` (units → uV, type → EEG / MISC)
- Handle and log unexpected edge cases

Dataset Preparation



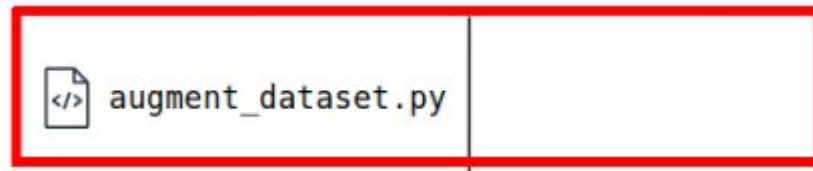
Dataset Preparation



augment_dataset.py

- Add status column to all *_channels.tsv files (good / bad)

Dataset Preparation



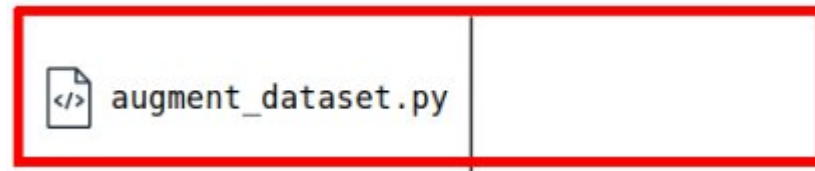
- Add status column to all `*_channels.tsv` files (good / bad)
- Add synthetic tasks
 - freeView = Symbol Search + Naturalistic Viewing
 - allTasks = all tasks of the subject

sub-<id>_task-symbolSearch_run-1_eeg.set
sub-<id>_task-ThePresent_run-1_eeg.set
sub-<id>_task-FunwithFractals_run-1_eeg.set
sub-<id>_task-DiaryOfAWimpyKid_run-1_eeg.set
sub-<id>_task-DespicableMe_run-1_eeg.set

↓ (symlink)

sub-<id>_task-freeView_run-1_eeg.set
sub-<id>_task-freeView_run-2_eeg.set
sub-<id>_task-freeView_run-3_eeg.set
sub-<id>_task-freeView_run-4_eeg.set
sub-<id>_task-freeView_run-5_eeg.set

Dataset Preparation



- Add status column to all `*_channels.tsv` files (good / bad)
- Add synthetic tasks
 - `freeView` = Symbol Search + Naturalistic Viewing
 - `allTasks` = all tasks of the subject

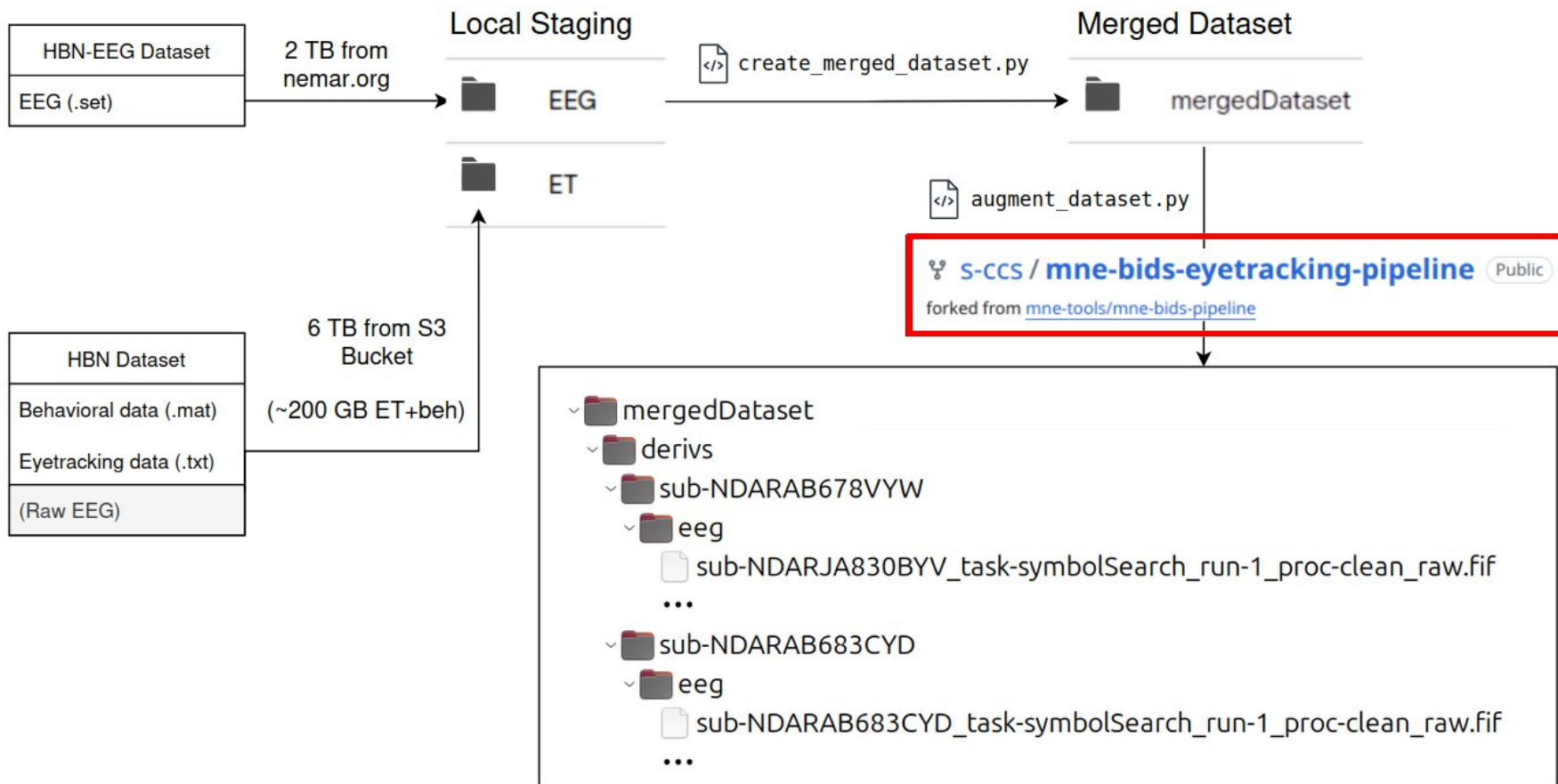
Data used for ICA fitting	Mean duration
Single movie recording	03:03 min
Symbol Search recording	03:42 min
Rule-of-thumb target	16:24 min
Synthetic freeView task	10:27 min
Synthetic allTasks task	30:13 min

sub-<id>_task-symbolSearch_run-1_eeg.set
sub-<id>_task-ThePresent_run-1_eeg.set
sub-<id>_task-FunwithFractals_run-1_eeg.set
sub-<id>_task-DiaryOfAWimpyKid_run-1_eeg.set
sub-<id>_task-DespicableMe_run-1_eeg.set

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sub-<id>_task-freeView_run-1_eeg.set
sub-<id>_task-freeView_run-2_eeg.set
sub-<id>_task-freeView_run-3_eeg.set
sub-<id>_task-freeView_run-4_eeg.set
sub-<id>_task-freeView_run-5_eeg.set

Dataset Preparation



Dataset Preparation



Raw .set file

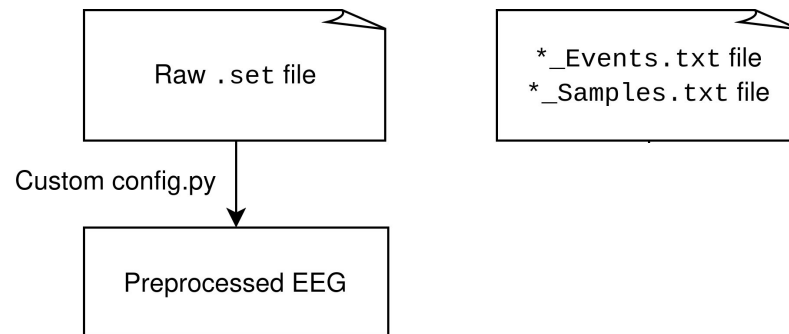
*_Events.txt file
*_Samples.txt file

l

Dataset Preparation



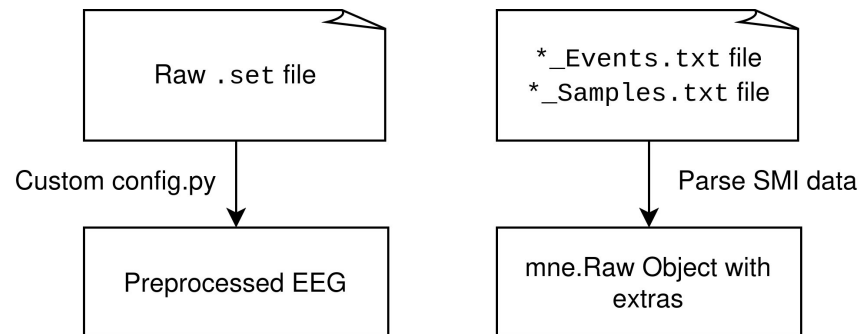
- EEG preprocessing (tailored to free viewing)



Dataset Preparation



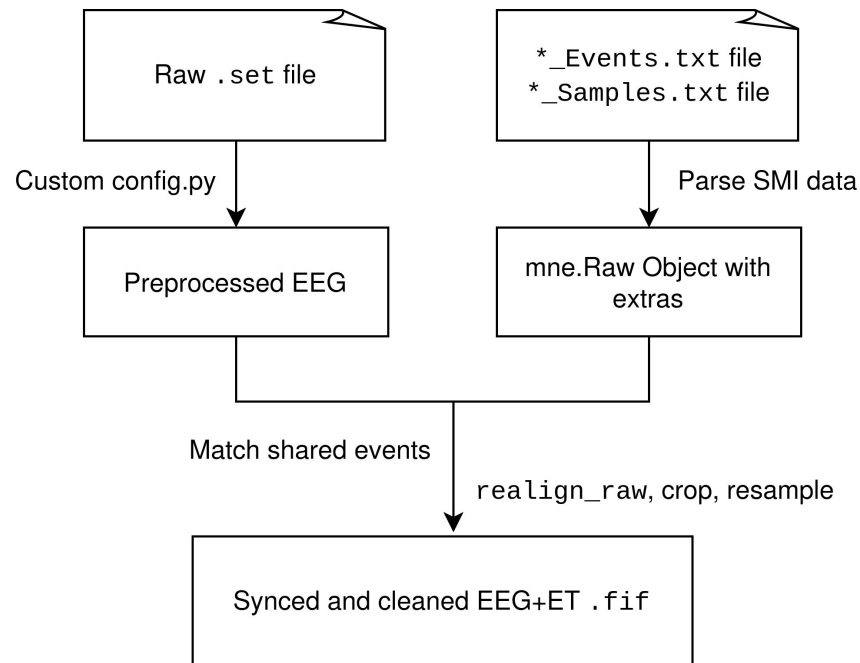
- EEG preprocessing (tailored to free viewing)
- Custom SMI eye-tracking parser



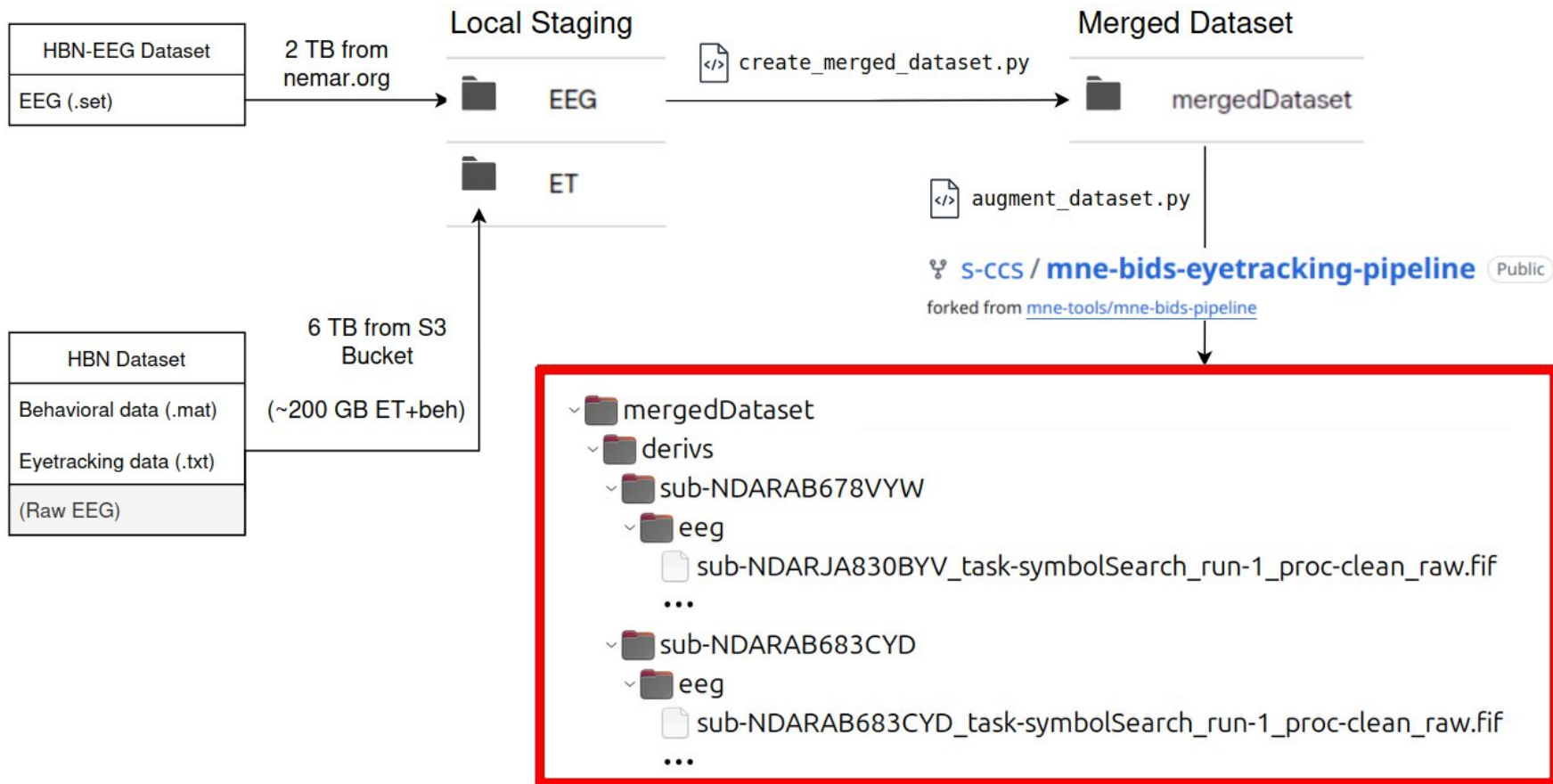
Dataset Preparation



- EEG preprocessing (tailored to free viewing)
- Custom SMI eye-tracking parser
- Robust matching of shared events



Dataset Preparation

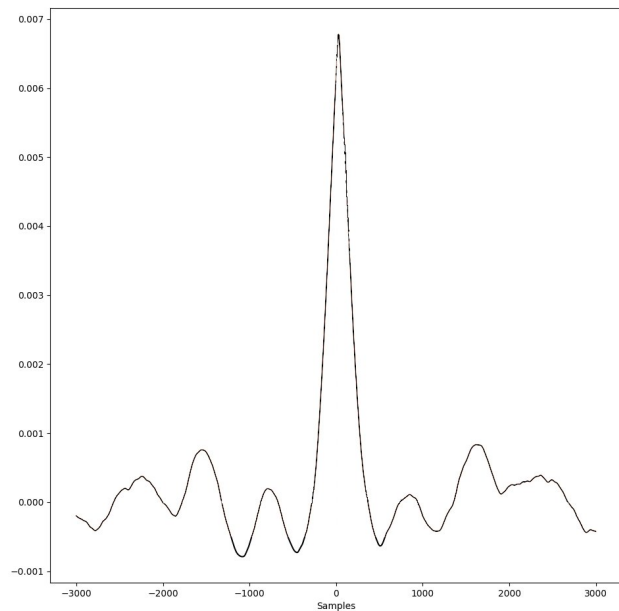


Structure

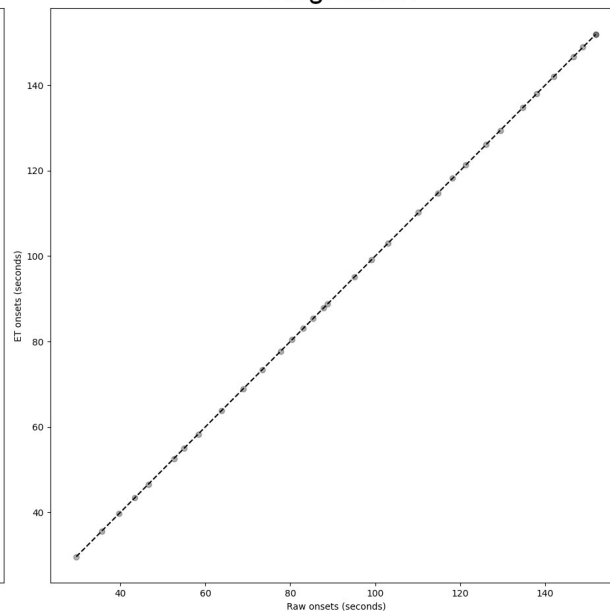
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Subject Curation

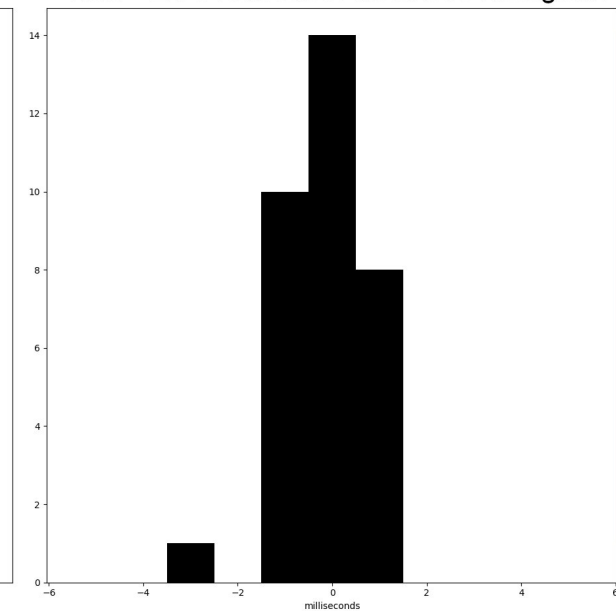
Cross correlation HEOG and ET

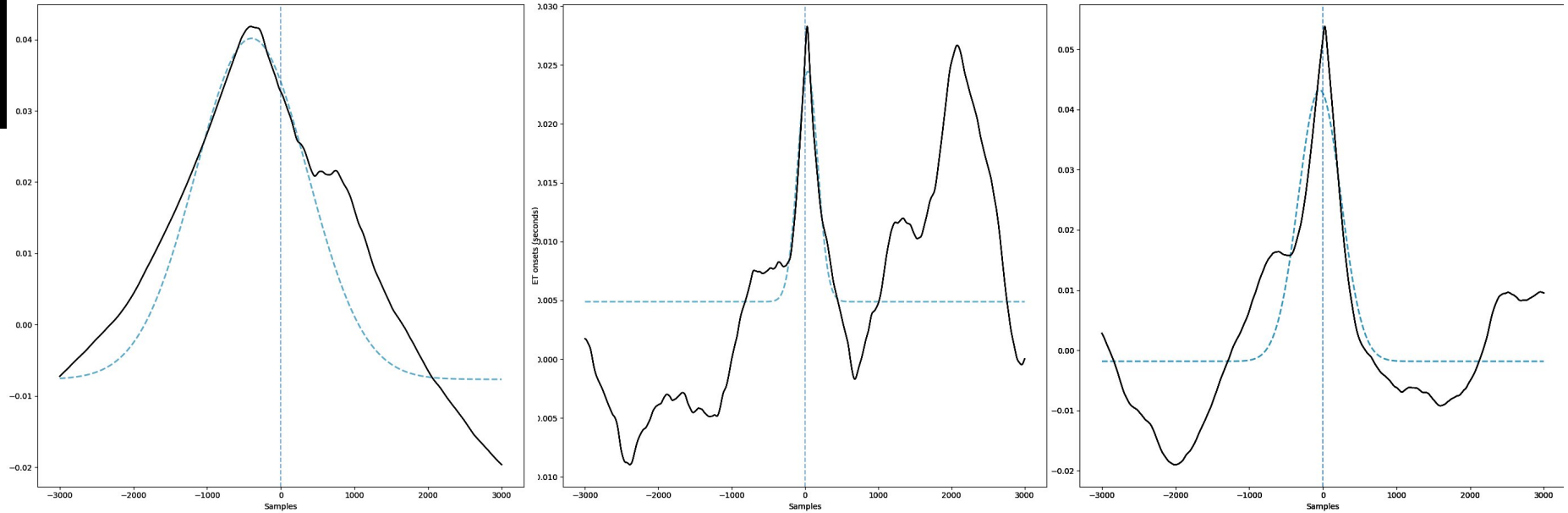


Regression



Raw - ET event onset distances histogram





Subject Curation

- Criteria for well synchronized recordings:

Criterion	Remaining	Share
All considered recordings	20317	100.00%
$ \mu_{\text{gauss}} \leq 200$	8491	41.79%
Cosine similarity ≥ 0.98	5711	61.41%
$ \text{xcorr peak idx} \leq 200$	3784	24.16%

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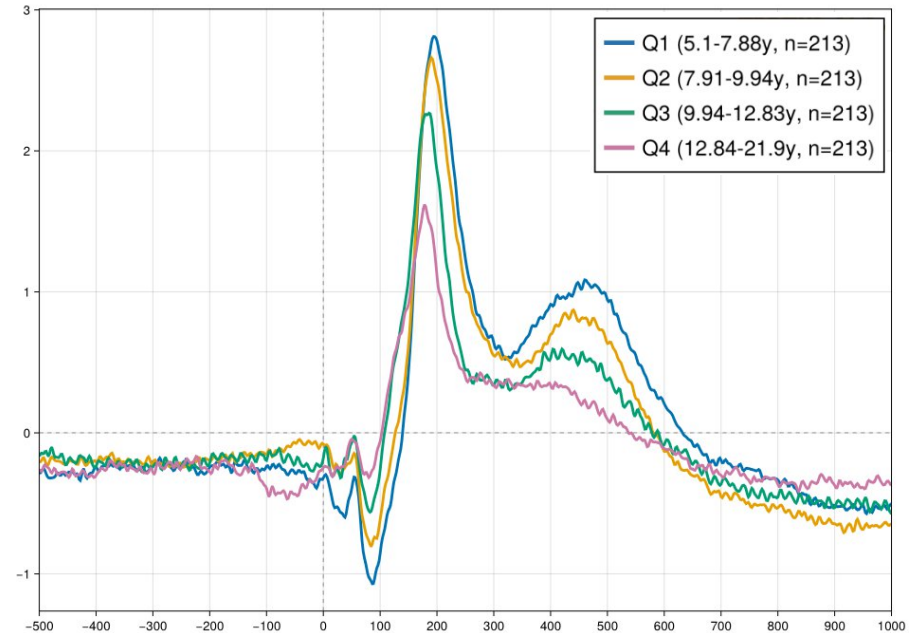
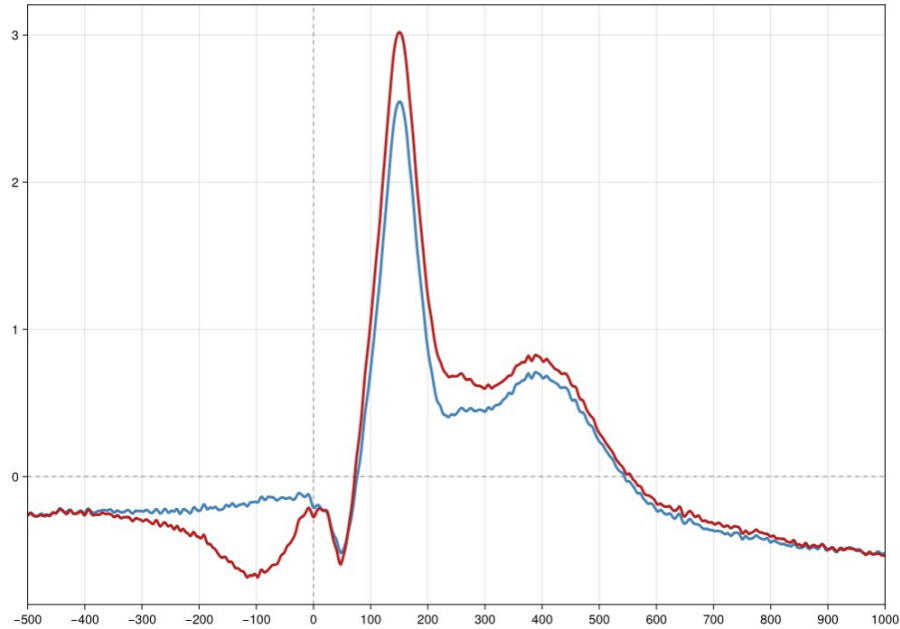
- Thesis uses subset without symbol search recordings
 - 876 subjects with 1355 recordings overall

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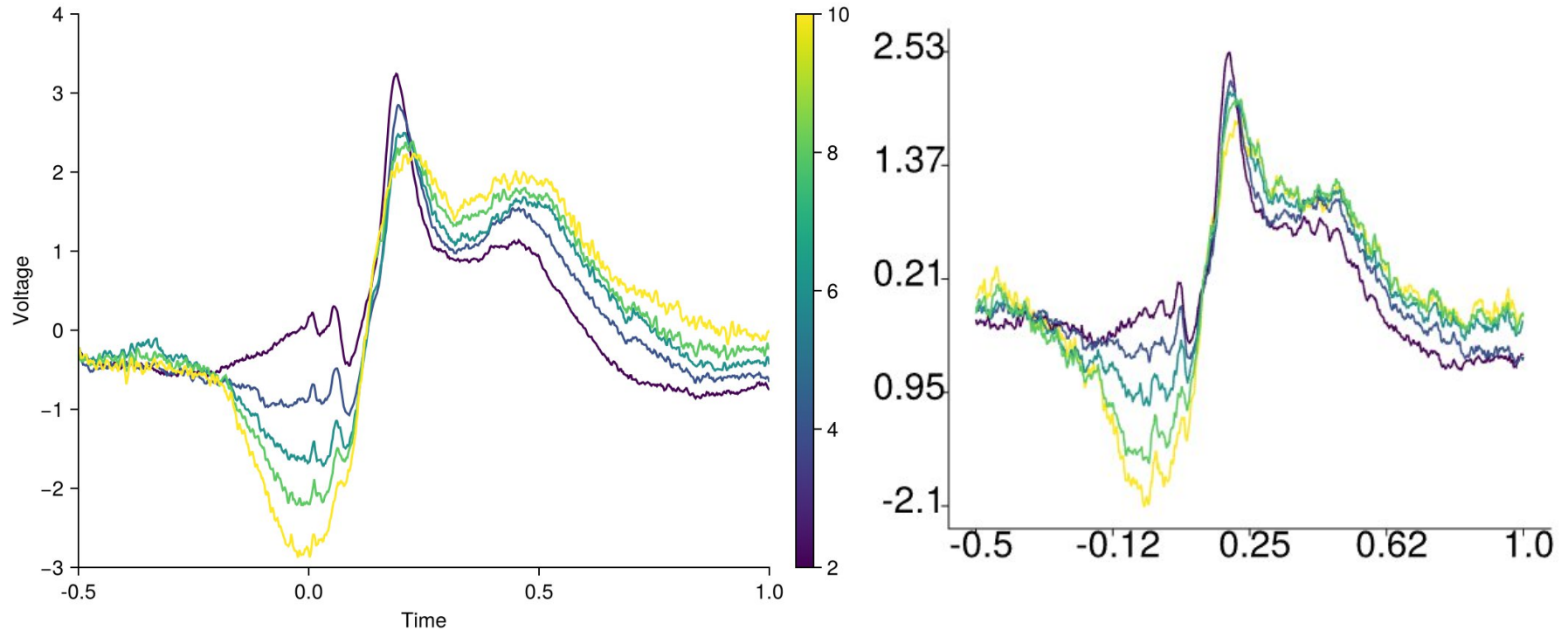
Analysis Results

- Uncorrected FRPs (red is unprocessed)



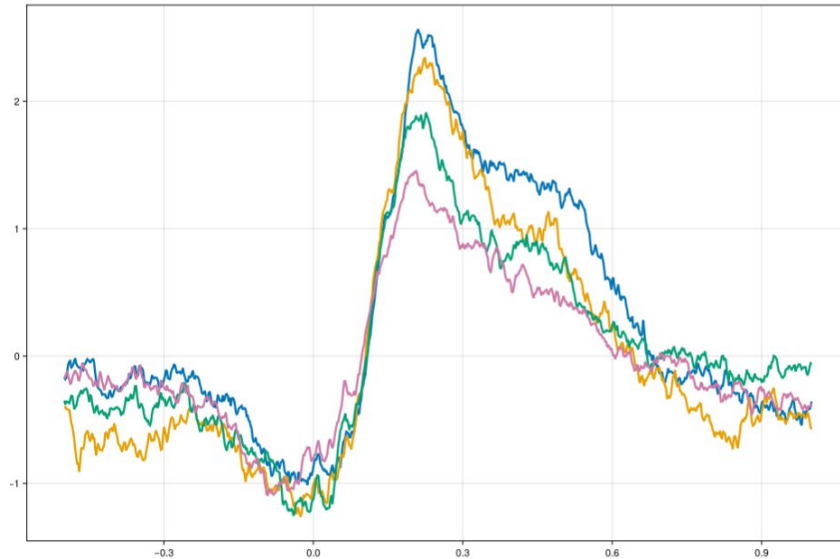
Analysis Results

- Corrected FRPs (left only uses „well synchronized“ recordings)



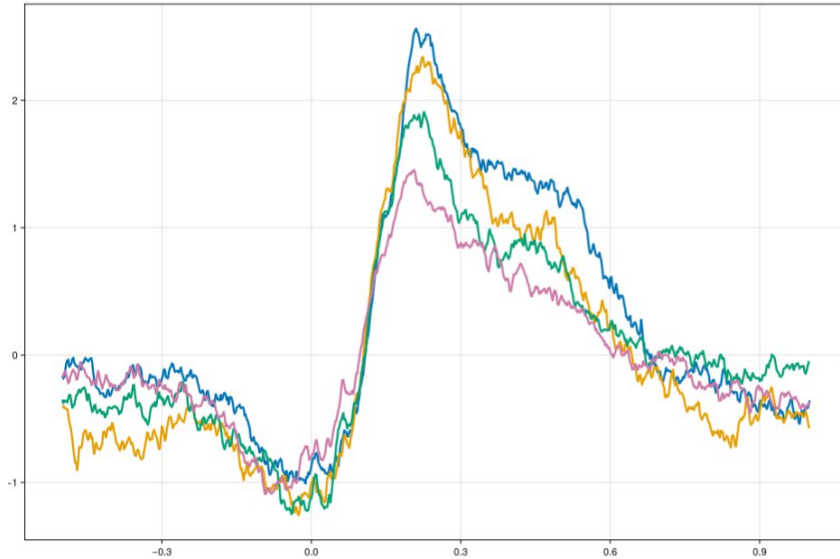
Analysis Results

- Amplitude 8 split by age

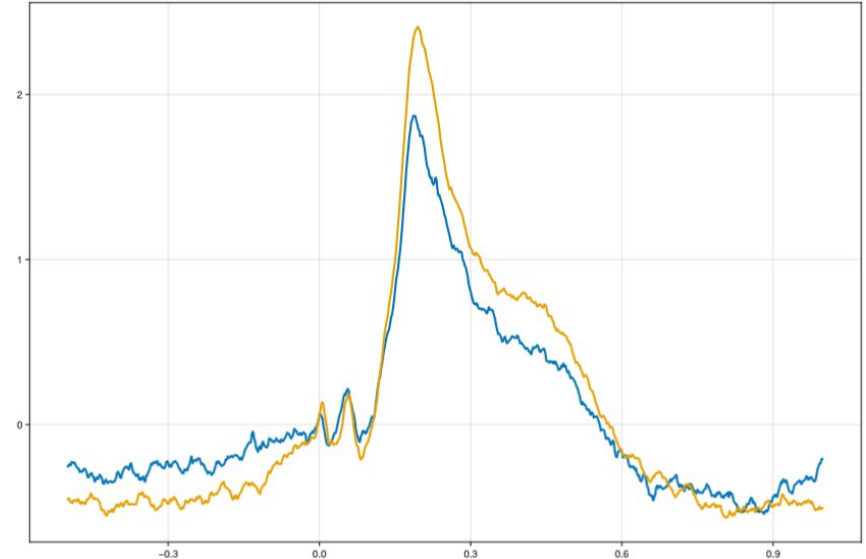


Analysis Results

- Amplitude 8 split by age



- Amplitude 2 split by sex (blue is female)

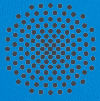


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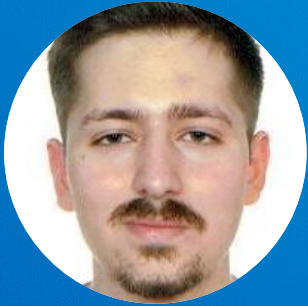
Takeaways

- HBN data can be unified into a usable EEG+ET dataset
- Synchronization quality impacts results, but scale helps
- FRP analysis yields interesting results, but preprocessing pipeline needs to be adjusted



University of Stuttgart
Institute for Visualization and Interactive Systems

Thank you!



Okan Mazlum

e-mail st156392@stud.uni-stuttgart.de

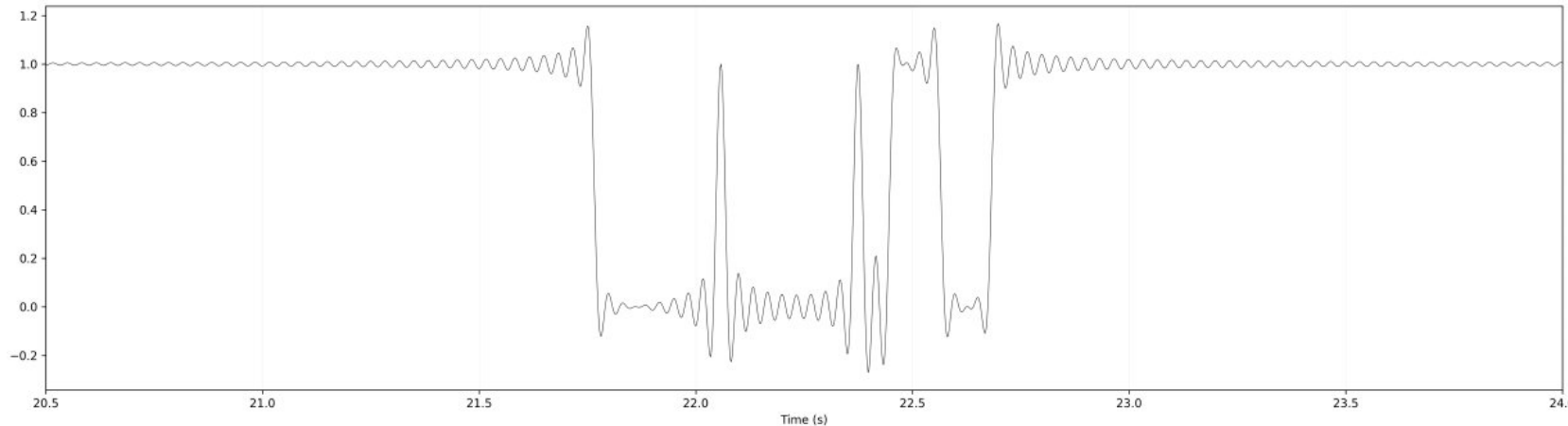
University of Stuttgart
Department of Computational Cognitive Science
Pfaffenwaldring 5a, 70569 Stuttgart

Dataset Preparation – Unique challenges

- 20 Hz noise in some recordings → custom PSD spike detection

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 - Fallback EOG electrodes

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- Fixed `raw.filter` only being applied to first run in ICA
- Many minor changes to make pipeline workable with multi-run dataset of this size

(Image) Sources

- Dimigen O, Ehinger BV. Regression-based analysis of combined EEG and eye-tracking data: Theory and applications. *J Vis.* 2021 Jan 4;21(1):3. doi: 10.1167/jov.21.1.3. PMID: 33410892; PMCID: PMC7804566.
 - Figures modified
- Alexander, L., Escalera, J., Ai, L. *et al.* An open resource for transdiagnostic research in pediatric mental health and learning disorders. *Sci Data* **4**, 170181 (2017). <https://doi.org/10.1038/sdata.2017.181>
- Seyed Yahya Shirazi, Alexandre Franco, Maurício Scopel Hoffmann, Nathalia B. Esper, Dung Truong, Arnaud Delorme, Michael P. Milham, Scott Makeig
bioRxiv 2024.10.03.615261; doi: <https://doi.org/10.1101/2024.10.03.615261>
 - Figures modified
- Langer, N., Ho, E., Alexander, L. *et al.* A resource for assessing information processing in the developing brain using EEG and eye tracking. *Sci Data* **4**, 170040 (2017). <https://doi.org/10.1038/sdata.2017.40>
- Kuefner, Dana & de Heering, Adélaïde & Jacques, Corentin & Palmero-Soler, Ernesto & Rossion, Bruno. (2010). Early Visually Evoked Electrophysiological Responses Over the Human Brain (P1, N170) Show Stable Patterns of Face-Sensitivity from 4 years to Adulthood. *Frontiers in Human Neuroscience*. 3. 67. 10.3389/neuro.09.067.2009.
 - Figures modified